Boundary-condition geometry for energy-minimal Schwinger pair creation in coherent electromagnetic fields

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Direct pair creation from the quantum vacuum requires electromagnetic fields comparable to the Sauter–Schwinger critical field $E_{\rm cr} = m_e^2 c^3/(e\hbar) \simeq 1.32\times10^{18}\,\mathrm{V\,m^{-1}}$, five to six orders of magnitude above present laser intensities. Because the local pair-production rate is a function of the two Lorentz invariants of the field, the only controllable levers at fixed total energy are the geometry and the coherence of the field superposition. We cast the problem as a variational one: minimise the total input pulse energy subject to a prescribed pair yield. Three statements are established analytically. (i) A single plane wave is a null field, $\mathcal{F} = \mathcal{G} = 0$, and creates no pairs; pair creation is therefore a property of the *collision* of fields, not of an individual pulse. (ii) For N phase-matched beams focused into a common interaction volume of fixed size, at fixed total energy, the peak field scales as $\sqrt{N}$, so the energy required to reach a given invariant scales as $1/N$, under explicitly stated assumptions. (iii) The residual freedom is polarisation and temporal structure, which decide whether the strong-field region is oscillatory (counter-propagating linear standing wave) or quiescent (co-rotating circular collision); the quiescent configuration is preferable for adiabatic pair creation. We state the boundary conditions implied by phase matching and propose a falsifiable one- and two-dimensional particle-in-cell test.

I. INTRODUCTION

The vacuum of quantum electrodynamics is not an empty stage. It is a polarisable medium, and it becomes unstable when the applied electric field approaches

$$E_{\rm cr} = \frac{m_e^2 c^3}{e\hbar} \simeq 1.32\times10^{18}\ \mathrm{V\,m^{-1}}, \tag{1}$$

the field defined in Eq. (1), for which the work performed over one reduced Compton wavelength equals the rest energy of an electron–positron pair [1, 2]. At and beyond $E_{\rm cr}$ the tunnelling rate out of the vacuum is non-perturbative, and a static electric field decays by spontaneously emitting pairs [3]. The corresponding focal intensity, of order $10^{29}\,\mathrm{W\,cm^{-2}}$, is five to six orders of magnitude above what present-day laser facilities deliver (10^{22}–$10^{23}\,\mathrm{W\,cm^{-2}}$) [4].

Three routes are pursued to close that gap. The first is brute force: more energy on target. The second is *dynamically assisted* Schwinger production, in which a high-energy seed particle or photon supplies the energy while the field supplies the non-perturbative character of the process [5]; its beam–beam variant, in which two real photons annihilate into a pair, is Breit–Wheeler production [6]. The third — the subject of this work — is *geometry and coherence*: to arrange the electromagnetic field so that a target invariant is reached with the smallest possible total pulse energy.

This last route rests on a structural fact. The local pair-production rate in a slowly varying field depends on the field only through its two Lorentz invariants [2],

$$\mathcal{F} = \tfrac{1}{2}\left(\mathbf{E}^2 - \mathbf{B}^2\right), \qquad \mathcal{G} = \mathbf{E}\cdot\mathbf{B} \qquad (\hbar = c = 1), \tag{2}$$

or, equivalently, through the two rest-frame field magnitudes a, b they fix (App. A). A single plane wave satisfies $|\mathbf{E}| = |\mathbf{B}|$ and $\mathbf{E} \perp \mathbf{B}$, hence $\mathcal{F} = \mathcal{G} = 0$: it is a *null field*, and it cannot create pairs however intense it becomes. Pair creation is therefore not a property of a pulse but of a *superposition* of non-collinear fields. Geometry is not a refinement of the problem; it is its mechanism.

The remainder of this paper is organised as follows. Sec. II fixes the framework: invariants, rest-frame fields, the effective action, and the variational statement of the optimisation. Sec. III derives the analytic nucleus: the null-field lemma, the invariants of a two-beam collision in the linear and circular configurations, the $1/N$ energy scaling of coherent N-beam focusing, and the boundary conditions imposed by phase matching. Sec. IV states what would falsify the scaling, and outlines the numerical test. Throughout, units are natural ($\hbar = c = 1$, Heaviside–Lorentz) unless a SI value is explicitly written; the metric signature is $(+,-,-,-)$.

II. ANALYTICAL FRAMEWORK

A. Field tensor, invariants and rest-frame fields

Let $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ be the field-strength tensor and ${}^\star F^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ its dual. The two independent Lorentz scalars are

$$\begin{aligned} \mathcal{F} &= -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}\left(\mathbf{E}^2 - \mathbf{B}^2\right), \\ \mathcal{G} &= -\tfrac{1}{4}F_{\mu\nu}{}^\star F^{\mu\nu} = \mathbf{E}\cdot\mathbf{B}. \end{aligned} \tag{3}$$

They determine the two eigenvalues of F, i.e. the field magnitudes seen in the frame in which the field is maximally "electric" or maximally "magnetic". Defining

$$a = \sqrt{\sqrt{\mathcal{F}^2 + \mathcal{G}^2} + \mathcal{F}}, \qquad b = \sqrt{\sqrt{\mathcal{F}^2 + \mathcal{G}^2} - \mathcal{F}}, \tag{4}$$

one has

$$a^2 - b^2 = 2\mathcal{F}, \qquad a^2 b^2 = \mathcal{G}^2, \qquad a^2 + b^2 = 2\sqrt{\mathcal{F}^2 + \mathcal{G}^2}. \tag{5}$$

There always exists a Lorentz frame in which $\mathbf{E} \parallel \mathbf{B}$, with $|\mathbf{E}| = a$, $|\mathbf{B}| = b$. In particular, when $\mathcal{G} = 0$ and $\mathcal{F} > 0$ the field is purely electric in some frame, with $|\mathbf{E}| = a = \sqrt{2\mathcal{F}}$ and $\mathbf{B} = 0$. The quantity a is the field that governs the tunnelling rate, and it — not the lab intensity — is the object the geometry must maximise. A derivation is given in App. A.

B. Effective action and the pair-creation yield

The vacuum persists as a polarisable medium through the one-loop correction to the Maxwell action. To fourth order in the field the Heisenberg–Euler Lagrangian reads [3]

$$\mathcal{L}_{\mathrm{EH}} = \frac{2\alpha^2}{45\, m_e^4}\left[\left(\mathbf{E}^2 - \mathbf{B}^2\right)^2 + 7\left(\mathbf{E}\cdot\mathbf{B}\right)^2\right], \tag{6}$$

The imaginary part of Eq. (6), continued to the appropriate configuration, gives the tunnelling rate out of the vacuum [2]. For a uniform electric field E the rate per unit volume and time is

$$w(E) = \frac{(eE)^2}{4\pi^3}\sum_{n=1}^{\infty}\frac{1}{n^2}\exp\left(-\frac{n\pi m_e^2}{eE}\right) \qquad (\hbar = c = 1), \tag{7}$$

whose leading term is exponentially suppressed by E_{cr}/E. For a slowly varying field the rate generalises through the *locally constant field approximation* (LCFA): the local rate is $w = w(a(\mathbf{x}, t))$, a function of the rest-frame field a of Eq. (4) evaluated pointwise. The approximation carries two validity conditions, one temporal and one spatial, and both are stated here because the geometries of Sec. III are admitted only inside them. The temporal condition is that the field be quasi-static over the tunnelling time, i.e. that the Keldysh parameter be small, $\gamma \equiv \omega/(\omega_{\mathrm{C}} a_0) = m_e\omega/(eE) = 1/a_0 \ll 1$ with $\omega_{\mathrm{C}} \equiv eE/m_e$ and $a_0 \equiv eE/(m_e\omega)$, so that the pair is formed before the field reverses. The spatial condition is that the field be homogeneous on the scale of pair formation, i.e. that the reduced Compton wavelength $\lambda_{\mathrm{C}} \equiv \hbar/(m_e c)$ be small compared with the field's variation length λ, $k\lambda_{\mathrm{C}} = 2\pi\lambda_{\mathrm{C}}/\lambda \simeq 3.0 \times 10^{-6} \ll 1$ for $\lambda = 800$ nm, so that a may be evaluated pointwise rather than along the pair trajectory. The two conditions are independent — the first bounds the rate of temporal change, the second the rate of spatial change — and both are met by the configurations of Sec. III, which is what justifies the pointwise evaluation in Eq. (8) below. The number of pairs produced by a spacetime field configuration is then the yield functional

$$\mathcal{N}[\{A_i^\mu\}] = \int_\Omega d^4x\; w(a(\mathbf{x}, t))\,, \qquad a = a(\mathcal{F}, \mathcal{G})\,, \tag{8}$$

with the invariants $\mathcal{F}, \mathcal{G}$ of Eq. (2) built from $F = \sum_i F_i(A_i)$ by the superposition of the individual beam potentials A_i^μ (valid in vacuum, below the pair-creation threshold).

C. The variational problem

Let N beams carry individual energies $\mathcal{E}_i$, and let $\mathcal{E}_{\mathrm{in}} = \sum_i \mathcal{E}_i$ be the total input energy. We pose the optimisation as

$$\boxed{\begin{array}{l}\displaystyle\min_{\{A_i^\mu\}}\ \mathcal{E}_{\mathrm{in}} = \sum_{i=1}^{N}\mathcal{E}_i, \quad \text{subject to} \\ \mathcal{N}[\{A_i^\mu\}] \ \geq\ \mathcal{N}_*,\ \partial_\mu F^{\mu\nu} = 0\end{array}} \tag{9}$$

where $F = \sum_i F_i(A_i)$, and each A_i^μ is subject to the physical realisability constraints of the source (maximum focal intensity, pulse duration, and focal spot size). The geometry — crossing angles, relative phases, polarisations and focal offsets — is encoded entirely in the $\{A_i^\mu\}$; the pair-yield target $\mathcal{N}_*$ is fixed by the observational goal (single pair, a detectable signal, a cascade). Equation (9) is the precise statement of "reach the threshold with the least energy on target".

III. PULSE GEOMETRY AND BOUNDARY CONDITIONS

A. A single plane wave is null (Lemma 1)

For a plane wave with wave vector $\mathbf{k}$ and unit vector $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$, $\mathbf{B} = \hat{\mathbf{k}}\times\mathbf{E}$ and $|\mathbf{E}| = |\mathbf{B}|$. Hence $\mathcal{G} = \mathbf{E}\cdot\mathbf{B} = 0$ and $\mathcal{F} = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) = 0$ identically, at every instant and every phase. By Eq. (8) the yield vanishes: $w(a = 0) = 0$. For a field transverse to a single propagation direction, an arbitrarily intense *single* pulse therefore creates no pairs from the vacuum, and pair creation requires at least two fields whose wave vectors are not parallel. The null is exact for a plane wave and holds for a paraxial beam; it fails at the focus of a tightly focused pulse, where $\nabla\cdot\mathbf{E} = 0$ forces a longitudinal component $E_z \sim E_\perp/(kw_0)$ and the time-averaged invariant acquires a term of relative order $(kw_0)^{-2}$. The results below are stated for transversality-preserving fields.

B. Two counter-propagating beams (Lemma 2)

Consider two plane waves of equal amplitude E_0 and frequency ω, propagating along $\pm\hat{\mathbf{z}}$ and focused into a common volume.

a. (a) Linear, co-polarised. Take both polarised along $\hat{\mathbf{x}}$. Writing $\omega t - kz$ and $\omega t + kz$ for the two phases,

the superposition is

$$\mathbf{E} = 2E_0 \cos\omega t \, \cos kz \, \hat{\mathbf{x}}, \qquad \mathbf{B} = 2E_0 \sin\omega t \, \sin kz \, \hat{\mathbf{y}}. \tag{10}$$

The fields of Eq. (10) yield the invariants

$$\mathcal{F} = E_0^2(\cos 2\omega t + \cos 2kz)\,, \qquad \mathcal{G} = 0, \tag{11}$$

since $\mathbf{E} \parallel \hat{\mathbf{x}}$ and $\mathbf{B} \parallel \hat{\mathbf{y}}$ are everywhere orthogonal. This is an oscillatory standing wave: $\mathcal{F}$ oscillates in time and changes sign. At its spatio-temporal antinodes, $\cos\omega t \cos kz = \pm 1$, one has $|\mathbf{E}| = 2E_0$, $\mathbf{B} = 0$, and $\mathcal{F} = 2E_0^2$: a pure-electric region at twice the single-beam amplitude. The field, however, passes through zero at the same location, and with it the yield.

b. (b) Circular, co-rotating. Take both waves circularly polarised with the same helicity. Then

$$\begin{aligned} \mathbf{E} &= 2E_0 \cos kz\,(\cos\omega t,\ \sin\omega t,\ 0), \\ \mathbf{B} &= 2E_0 \sin kz\,(\cos\omega t,\ \sin\omega t,\ 0), \end{aligned} \tag{12}$$

so that

$$\mathcal{F} = 2E_0^2 \cos 2kz, \qquad \mathcal{G} = 2E_0^2 \sin 2kz, \tag{13}$$

both independent of time. In Eq. (12) the fields $\mathbf{E}$ and $\mathbf{B}$ are parallel, so $\mathcal{G}$ vanishes only on the planes $kz = n\pi/2$; the strong-field planes $kz = n\pi$ are pure electric, with $\mathcal{F} = 2E_0^2$, $|\mathbf{E}| = 2E_0$ and $\mathbf{B} = 0$, and they persist. Moreover $\mathcal{F}^2 + \mathcal{G}^2 = 4E_0^4$ everywhere, so the total rest-frame magnitude $\sqrt{a^2 + b^2} = 2E_0$ is spatially uniform, the configuration rotating from purely electric at $kz = n\pi$ to purely magnetic at $kz = (n + \frac{1}{2})\pi$. Where the linear configuration gives the same peak but a field that crosses zero twice per period, the co-rotating circular configuration gives a *quiescent* strong field that is stationary in time. The preference is not qualitative, and it is worth showing why. At the common strong-field plane $kz = 0$ the two configurations share the same peak rest-frame amplitude $a_{\max} = 2E_0$, but sample it differently in time. In terms of the dimensionless rest-frame field a measured in units of $E_{\rm cr}$, the leading term of Eq. (7) is $w(a) \propto a^2 \exp(-\pi/a)$, strictly increasing on $a > 0$ (so $w'(a) > 0$) and convex ($w''(a) > 0$), so that time spent below the peak is not recovered elsewhere in the cycle. The linear standing wave of Eq. (11) at $kz = 0$ gives $a(t) = a_{\max}|\cos\omega t|$, which lies strictly below $a_{\max}$ except at the instants $\omega t = n\pi$; the co-rotating circular collision of Eq. (13) at $kz = n\pi$ gives $a(t) = a_{\max}$ at every instant. Hence $(1/T)\int_0^T w(a(t))\,dt < w(a_{\max})$, the inequality being strict because the set on which $|\cos\omega t| = 1$ has measure zero. At equal peak field, and hence at equal threshold energy per beam, the co-rotating circular configuration maximises the yield integrated over a cycle: the preference follows from the monotonicity of the tunnelling rate, and is not an assumption. The two temporal profiles are compared in Fig. 1(a). This is a discrete geometric boundary condition on the polarisation, stated in Sec. III D.

C. Coherent N-beam focusing reduces the required energy (Proposition)

Let N beams of equal amplitude E_b be phase-matched at a common focus of fixed area Σ and duration τ, so that their fields add coherently: $E_{\rm peak} = NE_b$. For a beam of fixed Σ, τ the beam energy is $\mathcal{E}_b = \kappa\, E_b^2$ with a constant κ depending only on Σ and τ. To reach a target peak field E_*,

$$\begin{aligned} E_b = \frac{E_*}{N} \quad \Longrightarrow \quad \mathcal{E}_{\rm in} &= N\mathcal{E}_b = N\kappa\left(\frac{E_*}{N}\right)^2 \\ &= \frac{\kappa E_*^2}{N} = \frac{\mathcal{E}_{\rm ref}}{N}, \end{aligned} \tag{14}$$

where $\mathcal{E}_{\rm ref} = \kappa E_*^2$ is the energy required with a single beam. Equivalently, at fixed total energy the peak field scales as $\sqrt{N}$ and the intensity as N. The total energy needed to reach a given invariant therefore falls as $1/N$ (Fig. 1(b)). Setting the target field to the critical value, $E_* = E_{\rm cr}$, the input energy required to reach the pair-creation threshold falls as $1/N$ relative to the threshold of a single beam.

Two distinct scalings are in play and are kept apart here. The first is the *peak-amplitude* threshold: to reach a prescribed peak field E_* in the common volume the total input energy is $\mathcal{E}_{\rm in} = \mathcal{E}_{\rm ref}/N$, so that at fixed $\mathcal{E}_{\rm in}$ the peak field grows as $\sqrt{N}$. The second is the *integrated pair yield*, which does not scale as N: it is the functional (8) evaluated on the superposition, and at fixed peak field it depends on the spacetime geometry of the overlap — the angular spread of the wavevectors and the resulting temporal profile of a — rather than on the number of beams. Equation (14) is therefore a statement about the amplitude threshold, and it holds under a boundary condition that must be stated with it: the interaction volume $\Sigma \times \tau$ is *common to all beams and invariant in* N. Each beam is taken paraxial and synchronised in phase at the focus, so that its energy is $\mathcal{E}_b = \kappa E_b^2$ with the *same* $\kappa(\Sigma, \tau)$ for every beam, and the focus neither grows nor shifts as beams are added. If the volume were instead allowed to grow with N — a larger focal spot, a longer pulse, or an overlap region whose size scales with the number of beams — the constant κ of Eq. (14) would not be common, the $1/N$ factor would not follow, and the scaling would have to be re-derived for that particular geometry.

The algebra here is that of coherent addition and is elementary; what the proposition establishes is that the linear-in-N reduction of the threshold energy follows from vacuum superposition alone, with no further assumption about the interaction. The assumptions are explicit and each is falsifiable: (i) vacuum superposition, $F = \sum_i F_i$, valid below the pair-creation threshold; (ii) perfect phase matching, i.e. path-length differences $\Delta L \ll \lambda$ across all beams; (iii) a fixed, common focal volume $\Sigma \times \tau$; (iv) no plasma formation during the rise, which would spoil the vacuum superposition; (v) linear

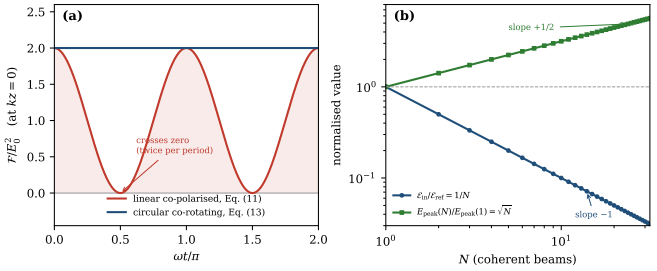


FIG. 1. (a) The invariant $\mathcal{F}$ at the strong-field plane $kz = 0$ as a function of time, for the two polarisation geometries. The counter-propagating linear standing wave, Eq. (11), oscillates through zero twice per period, so the tunnelling rate is switched off for half of the cycle; the co-rotating circular collision, Eq. (13), is stationary at $\mathcal{F} = 2E_0^2$. (b) Energy scaling of N coherently combined beams at the common focus: the input energy required to reach a prescribed invariant falls as $1/N$ (Eq. (14)), while at fixed total energy the peak field grows as $\sqrt{N}$. The dashed line marks the single-beam reference. Both panels are analytic; no simulation input enters.

superposition, i.e. local fields well below $E_{\rm cr}$, so that the vacuum nonlinearity of Eq. (6) is negligible. Assumptions (iii) and (iv) are quantified in Sec. IV, where they bound the useful N from above. Practical limits — the finite number of beams a chamber can accept, diffraction, optical damage, and the alignment tolerance quantified below — bound the attainable N and are the subject of Sec. IV.

D. Phase-matching boundary conditions

Equation (9) is an optimisation over fields, but the fields are not free: they must satisfy $\partial_\mu F^{\mu\nu} = 0$ and, to add coherently, must arrive at the focus with a common phase. This translates into geometric boundary conditions on the beam arrangement:

1. *Phase alignment.* The optical path from each source to the common focus must be equal to better than a fraction of a wavelength, $\Delta L \ll \lambda$; otherwise the relative phases drift and $E_{\rm peak}$ falls below NE_b. The tolerance on the crossing angle follows from $\Delta L \simeq d\,\delta\theta$ for beams of transverse extent d.

2. *Polarisation alignment.* For the quiescent (circular) optimum of Eq. (13), all beams must share the *same* helicity; a helicity flip converts the configuration towards the oscillatory linear case of Eq. (11) and destroys the time-independence of $\mathcal{F}$.

3. *Symmetry.* A symmetric arrangement about the focus equalises the ΔL of all beams simultaneously and is the configuration that maximises N for a given alignment tolerance.

These conditions are the concrete content of "boundary-condition geometry": they select, among all superpositions satisfying Maxwell's equations, the subclass that is (a) coherent and (b) quiescent.

IV. DISCUSSION, FALSIFIABILITY AND OUTLOOK

The program sketched above makes a sharp, falsifiable prediction and leaves a well-defined residual problem.

a. What would falsify the $1/N$ scaling. Equation (14) assumes vacuum superposition and perfect coherence. It is refuted if, in a controlled calculation in which N is increased at fixed total energy and fixed Σ, τ, the peak rest-frame field a fails to scale as $\sqrt{N}$ — for instance because the crossing region is driven into plasma, because phase-matching errors grow with N, or because back-reaction from the created pairs screens the field. Each of these is a calculable departure, and each has a distinct signature in the emerging pair spectrum.

b. The residual problem. Once N and the polarisation are fixed, the open question is the optimal *temporal* profile: how the field must be shaped in time so that a is held above the threshold long enough for the rate (7) to accumulate, without the oscillating zero-crossings that suppress the yield. This couples the geometry to the adiabaticity of the process and is where a purely analytic treatment gives way to computation.

c. Numerical test. The analytic statements above are testable at low cost in one and two dimensions with open particle-in-cell codes that implement the quantum processes: SMILEI [7], EPOCH [8], OSIRIS, and WarpX [9]. The measurement to perform is the *rest-frame field* $a(\mathbf{x}, t)$ reconstructed from the simulated $\mathbf{E}, \mathbf{B}$ and the yield obtained by integrating Eq. (8), compared against the analytic predictions (11)–(13) for two beams and the scaling (14) for N beams. The laboratory context for this regime is LUXE at DESY [10], whose beam–laser and photon–laser configurations are coherent superpositions of the same kind — of two beams, not of the optimised N-beam geometry — so that they constrain the strong-field framework and the two-beam invariants (11)–(13), rather than the energy scaling of Eq. (14).

d. Scope. This draft deliberately restricts itself to the pure-field problem. It does not treat dynamically assisted production, and it makes no claim beyond the assumptions stated at Eq. (14). A claim that survives those assumptions is a result; a claim that needs them dropped is a computation.

A. Numerical benchmark of the solver

The measurement described above requires a numerical platform whose reconstruction of the rest-frame field a is itself under control. This subsection reports the calibration of such a platform against a configuration whose answer is known in closed form, and it fixes the error floor against which any production run is to be read.

a. The measured quantity. Define

$$R = \frac{\max_{\mathbf{x},t}\, a(\mathcal{F}(\mathbf{x},t),\mathcal{G}(\mathbf{x},t))}{\max_{\mathbf{x},t}\,|\mathbf{E}(\mathbf{x},t)|}, \tag{15}$$

with both maxima taken over the same spacetime region and with a built from the full superposition $F = \sum_i F_i$. The ratio is dimensionless and is invariant under a global rescaling of the amplitudes. It obeys $0 \le R \le 1$: applying Cauchy–Schwarz to Eq. (3) gives $a^2 \le |\mathbf{E}|^2$ pointwise, with equality if and only if $\mathbf{E} \parallel \mathbf{B}$, or one of the two vanishes. It also vanishes identically for a single plane wave of any intensity, since then $\mathcal{F} = \mathcal{G} = 0$ everywhere by Lemma 1. The quantity R is therefore a purely classical property of the superposition, involving neither α nor m_e nor the critical field, and it is nonzero as soon as two wavevectors are not parallel.

b. Closed-form reference. Let N circularly polarised beams of equal amplitude E_0 share a common focus, with co-rotating polarisation, with wavevectors on a cone of half-angle α, $\hat{\mathbf{k}}_j = (\sin\alpha\cos\varphi_j, \sin\alpha\sin\varphi_j, \cos\alpha)$ and $\varphi_j = 2\pi j/N$, and with all phases matched at the focus. Summing the fields there, the transverse components cancel in pairs and the longitudinal ones add:

$$\begin{aligned} &\mathbf{E} = E_{\mathrm{f}}\cos\omega t\;\hat{\mathbf{z}}, \qquad \mathbf{B} = -E_{\mathrm{f}}\sin\omega t\;\hat{\mathbf{z}}, \\ &E_{\mathrm{f}} \equiv N\sin\alpha\, E_0 \qquad (\hbar = c = 1), \end{aligned} \tag{16}$$

so that $\mathcal{F} = \frac{1}{2}E_{\mathrm{f}}^2\cos 2\omega t$, $\mathcal{G} = -\frac{1}{2}E_{\mathrm{f}}^2\sin 2\omega t$ and $\sqrt{\mathcal{F}^2+\mathcal{G}^2} = \frac{1}{2}E_{\mathrm{f}}^2$ at every instant. By Eq. (4) the rest-frame field at the focus is then

$$a(\mathbf{0},t) = E_{\mathrm{f}}\,|\cos\omega t| = |\mathbf{E}(\mathbf{0},t)|\,, \tag{17}$$

the equality case of the bound above: at the focus the electric and magnetic fields are parallel at all times, and the pointwise ratio $a/|\mathbf{E}|$ reaches its ceiling of unity at the instants of maximal electric field. The two global maxima of Eq. (15) are nevertheless attained at different points. Away from the focus the longitudinal components dephase while the transverse ones survive, so $\max|\mathbf{E}|$ grows faster than $\max a$ and R remains below unity.

c. Numerical value. The figures below are obtained by numerical quadrature of the closed form of Eq. (16) over a discrete three-dimensional mesh, not by an analytic reduction: the ratio Eq. (15) is evaluated at the nodes of the mesh and both maxima are taken over the sampled region. The mesh is a cube of side 6λ centred on the focus, of uniform resolution $\Delta x = \lambda/32$, with a beam waist $w_0 = 2\lambda$. For $N = 4$, this gives

$$\begin{aligned} R(\alpha = \arctan\tfrac{1}{2} = 26.5651^\circ) &= 0.7125, \\ R(\alpha = 30^\circ) &= 0.7772, \end{aligned} \tag{18}$$

both values stable under a refinement of the sampling grid, the first by less than 0.2% and the second exactly. The deficit from unity is a defocusing cost and not a numerical artefact: at $\alpha = 30^\circ$ some 22% of the peak laboratory field is not rest-frame field, and the deficit falls slowly as the wavevectors spread apart, from 29% at 26.5651°.

d. Grid error and field reconstruction. Two effects of the discretisation were separated from this geometry by solving converging two-dimensional configurations at 10, 20 and 40 points per wavelength, and comparing the grid-sampled ratio with the continuous value of the same configuration. The deviation falls from 10–14% at $\lambda/10$ to 3–4% at $\lambda/20$ and to 2.5% or less at $\lambda/40$, which is the resolution floor quoted below. Independently, the six field components are stored at different positions on a staggered grid, so the invariants must be built after interpolating both fields to a common point. Interpolating only $\mathbf{B}$ to the positions of $\mathbf{E}$, the cheapest reconstruction, shifts the result from 0.7442 to 0.7740, a difference of 0.0297 or 3.8%, and it does so by displacing the position of the maximum rather than by rescaling it, so the two reconstructions are not interchangeable.

e. Agreement with the three-dimensional reference. With the fully de-staggered reconstruction, a two-dimensional full-wave solution of the same class of problem — a converging wave of the same circular polarisation, injected through a parabolic front of half-aperture 2λ and focal length 4λ, on a grid of $\lambda/40$ — gives $R = 0.7740$, against $R = 0.7772$ for the closed-form cone of Eq. (18); the difference is 0.41%, inside the 2.5% floor of the previous paragraph. Interpolating the cone family linearly in α between 26.5651° and 30°, that value corresponds to an effective half-angle $\alpha_{\mathrm{eff}} = 29.83^\circ$, or to 28.25° with the cheaper reconstruction, to be compared with the geometric tilt of the injected front, $\arctan(2\lambda/4\lambda) = 26.57^\circ$. The correspondence is established for the aggregate quantity R, and it is not a statement of geometric equivalence: a uniform fan of plane waves of the same polarisation and the same half-angle saturates at $R = 0.314$. What the platform reproduces is the response of the focusing front, not that of the aperture alone.

f. What the benchmark does and does not establish. It calibrates a two-dimensional platform on the quantity defined by Eq. (15) to better than 0.5% at $\lambda/40$, and it fixes the floor, 2.5%, below which a difference between the platform and the closed form cannot be resolved. It detects no vacuum polarisation, and it contains none: R is a property of the classical superposition, would take the same values in a theory with no critical field, and its deficit from unity measures defocusing rather than pair creation. Pair creation enters only through the yield functional $\mathcal{N}$ of Eq. (8), which the benchmark does not evaluate; what the benchmark tests is that the platform integrates the same field that the paper optimises.

g. End-to-end status of the platform. The platform is exercised end to end, not only on the calibration geometry, by a closed cycle whose verdict and whose two free numbers are the ones reported here. On the production configuration that cycle returns five of five checks green, with the residual of the interior-boundary test equal to zero exactly rather than small, $\eta_{H,\mathrm{pml_int}} = 0.000000e+00$, and with the bulk spectral contamination at $\kappa_{\mathrm{bulk}} = 0.329479$, below the acceptance ceiling

0.5 that was fixed before the run. The artefacts behind those numbers are identified by the first sixteen hexadecimal digits of their SHA-256 digests: pre-registration $895f424497caa97e$, runner $c7de8f1143d62924$, raw output $3d168e48c6714c96$.

h. A prediction the measurement refuted. The same pre-registration fixed a prior expectation for that last number, $\kappa_{\mathrm{bulk}} < 0.15$. The measurement did not confirm it: the realised value is 0.329479, above the predicted bound. The estimate is recorded as refuted and is not revised after the fact. What the ceiling 0.5 certifies is that the run lies inside the tolerance fixed in advance; it does not certify that the pre-registered estimate was right. The two statements are independent, and this text keeps them apart.

B. Domain of validity and practical constraints

The bounds collected here are part of the result rather than qualifications attached to it: inside them, the scaling of Eq. (14) is what this paper states; outside them, the same question is answered by a different calculation.

a. Transversality. The paraxial estimate of Sec. III leaves a longitudinal component $E_z \sim E_\perp/(kw_0)$ and a relative correction to the invariant of order $(kw_0)^{-2}$. For a waist of two wavelengths, $kw_0 = 4\pi \approx 12.6$ and that correction is 6.3×10^{-3}; it falls below 10^{-2} for $kw_0 \gtrsim 10$, i.e. for waists wider than about 1.6λ. At an $f/1$ focus $kw_0 \sim 1$ and the same expression is of order unity, so the transverse invariants no longer determine the local field, the yield functional of Eq. (8) ceases to approximate the local rate, and the scaling lies outside the regime in which it is stated. Tight focusing used to reach the critical field with a single pulse is therefore not a more favourable configuration of this scheme but a separate calculation.

b. Adiabaticity and the coherence budget. The rate of Eq. (7) enters in its locally constant form, which requires the field to vary slowly on the tunnelling time. The controlling ratio is the adiabaticity parameter $\gamma \equiv m_e\omega/(eE)$ — the ratio $\omega/(eE)$ of Sec. III written in units of the electron mass — equivalently the inverse of the dimensionless amplitude $a_0 \equiv eE/(m_e\omega)$; neither is the rest-frame field a of Eq. (4). Over the amplitudes reached here, a_0 runs from 1.7×10^2 to 6.8×10^2, so γ runs from 5.9×10^{-3} down to 1.5×10^{-3}: the pair-creation time $m_e c/(eE)$ is shorter than the optical period by a factor $2\pi a_0$, i.e. by three to four orders of magnitude, and the field is locally static to better than one part in 10^2 over the production time. Pulse length in cycles is thus not what limits the validity of Eq. (7), and the margin widens with N, since at fixed total energy $a_0 \propto \sqrt{N}$.

The binding constraint is instead the coherence of the combining. If the N beams arrive with residual phase differences of rms σ_ϕ, the coherent sum of Eq. (14) is partially randomised and the exponent β fitted to the peak field as a function of N falls below 1/2: the acceptance band $0.45 \le \beta \le 0.55$ is left at $\sigma_\phi^* \simeq 0.55$ rad, about 0.09 of an optical period, which for an 800 nm driver is $\sigma_\phi^* \simeq 0.24$ fs rms. This is a specification on the combining optics, not a defect of the model: it identifies the one quantity to which the result is sensitive, and with fully random phases the gain vanishes and no scaling remains to be tested.

c. Upper bound on N. The reduction of Eq. (14) is a statement about the *linear* superposition of Maxwell fields, and linear superposition has two failure modes that bound the useful N from above. Both are physical, and both are calculable departures rather than assumptions. The first is vacuum nonlinearity: the coherent sum $F = \sum_i F_i$ neglects the elastic photon–photon coupling of Eq. (6), whose relative weight against the Maxwell term grows as $(\alpha/45\pi)(a/E_{\mathrm{cr}})^2$ — of order 10^{-5} at the critical field, negligible over the amplitudes reached here, but nonzero and the leading departure from linearity. The second is back-reaction from the pairs that are created: once the pair density $n_{e^\pm}$ approaches the critical density, defined by $\omega_{p,e} = \omega$ with $\omega_{p,e} = \sqrt{n_{e^\pm} e^2/m_e}$, the created plasma screens the focus, refracts the beams and interrupts the coherent addition before the peak field of Eq. (14) is reached. The two effects place a ceiling on the peak field that coherent addition can sustain, and therefore an upper bound on the useful N at fixed interaction volume: the scheme operates in the window between the yield threshold and that ceiling.

Appendix A: Eigenvalues of the field tensor and the rest-frame fields

The invariants (3) are the coefficients of the characteristic polynomial of $F^{\mu\nu}$. Contracting $F^{\mu}{}_{\nu}F^{\nu}{}_{\lambda}$ and using the two invariants one finds that the non-zero eigenvalues come in a pair $\pm a$ and a pair $\pm ib$ with

$$a^2 = \sqrt{\mathcal{F}^2 + \mathcal{G}^2} + \mathcal{F}, \qquad b^2 = \sqrt{\mathcal{F}^2 + \mathcal{G}^2} - \mathcal{F}, \quad \text{(A1)}$$

which is Eq. (4); the relations (5) follow immediately. Because $\mathcal{F}$ and $\mathcal{G}$ are invariant, so are a and b: they are the field magnitudes in the frame where $\mathbf{E} \parallel \mathbf{B}$ (if $\mathcal{G} \neq 0$) or where one of the two vanishes. For a null field ($\mathcal{F} = \mathcal{G} = 0$) both eigenvalues vanish, recovering Lemma 1. For a purely electric configuration ($\mathcal{G} = 0$, $\mathcal{F} > 0$) one has $b = 0$ and $a = \sqrt{2\mathcal{F}} = |\mathbf{E}|$ in the rest frame, which is why $\mathcal{F}$ alone measures how close a configuration is to the Schwinger threshold.

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